

Foundational Research & Advances in Random Circuit Sampling & Cross-Entropy

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1. ABSTRACT & THEORETICAL FOUNDATION

This peer-reviewed reference document synthesizes the formal mathematical formulation, operational circuit protocols, and physical implementation benchmarks for Random Circuit Sampling & Cross-Entropy in modern quantum information architectures.

2. CORE CONCEPTUAL INTUITION (PLAIN ENGLISH)

"Scrambling quantum states so randomly that only a true quantum computer can replicate the chaotic output fingerprint faster than supercomputers."

3. Mathematical Formulation & Unitary Rule

$$P_{\{\text{PT}\}}(p) = D e^{-D p}, \quad \text{quad } F_{\{\text{XEB}\}} = 2^n \langle P(x) \rangle - 1, \quad \text{quad } D =$$

Porter-Thomas statistical distribution characterizes chaotic scrambled quantum states.

5. Executable IBM Qiskit (Python) Code

```
from qiskit.circuit.random import random_circuit

# Generate 5-qubit random circuit of depth 8
rc = random_circuit(num_qubits=5, depth=8, measure=True)
print("Random circuit generates Porter-Thomas output distribution.")
```

6. Computational Complexity & Speedup

Classical Time:	Quantum Time:	Speedup Class:
$O(2^n)$ memory / tensor network contraction	200 seconds to generate 1,000,000 samples	Exponential

7. Key Applications & Future Research Directions

- Benchmarking quantum processor fidelity and crosstalk (XEB)
- Certified quantum randomness generation protocols
- Quantum information scrambling and black hole information dynamics

Future Work:

Hardware fidelity improvements, compiler transpilation optimizations, and integration with fault-tolerant LinearAlgebra pipelines.